

Year 12 to Year 13 Transition Work

Nuclear Power Research Guide

The aim of this activity is to develop an understanding of how a modern nuclear power station operates. While watching the recommended video, produce your own notes covering the topics below.

You should not simply copy information from the video. Instead, aim to summarise the key ideas in your own words.

The online quiz will assess your understanding of the following topics and subtopics. Make notes accordingly.

1. Nuclear Fuel

- The fuel used in a modern nuclear reactor.
- The structure of a fuel assembly.
- Fuel rods and uranium fuel pellets.
- Why nuclear fuel is considered highly energy dense.

2. Nuclear Fission

- What happens during nuclear fission.
- The role of neutrons.
- How energy is released.
- How a nuclear chain reaction is maintained.

3. The Reactor

- The purpose of the reactor core.
- The function of the coolant.
- Why pressurised water reactors use separate primary and secondary water systems.
- The engineered barriers that prevent radioactive material escaping.

4. Heat Transfer

- How heat is transferred away from the reactor core.
- The purpose of the steam generator.
- Why the reactor must continue to be cooled after shutdown.

5. Electricity Generation

- How steam is produced.
- The purpose of the turbine.
- The purpose of the electrical generator.
- How the steam is recycled after passing through the turbine.

6. Radiation

- The difference between fresh fuel and spent fuel.
- Why spent fuel remains radioactive.
- How radiation can affect living tissue.
- Why iodine-131 is considered particularly hazardous after a nuclear accident.

7. Radioactive Waste

- What happens to spent nuclear fuel.
- How high-level radioactive waste is stored.
- Why the total volume of nuclear waste is relatively small.

8. Reactor Safety

- The purpose of the containment building.
- The meaning of “defence in depth”.
- Why refuelling operations require strict procedures.

9. The Nuclear Fuel Cycle

- How fresh fuel arrives at a power station.
- How fresh fuel is prepared before entering the reactor.
- What happens during a routine refuelling outage.

10. Nuclear Power in Context

- Why nuclear power is considered an efficient method of generating electricity.
- Advantages of modern reactor designs.
- The overall process by which a nuclear power station converts nuclear energy into electricity.